# GWENDAL LE COGUIC

# COURSES

Each training session focuses on practical demonstrations, interactive exercises, and discussions based on real-life situations to ensure the acquisition of key skills.

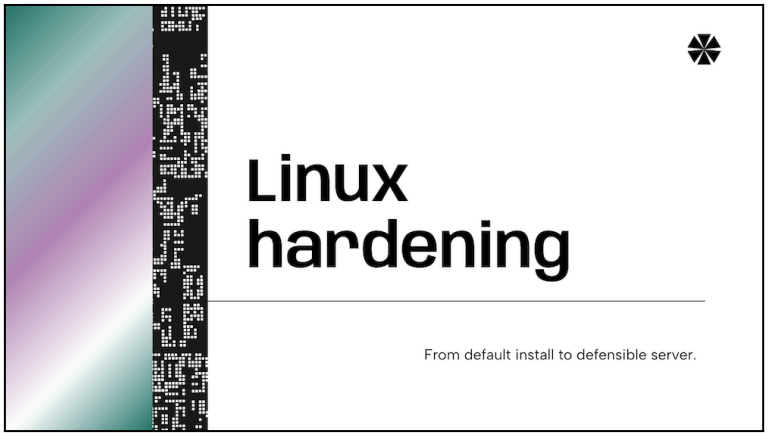
Each session can be taken:

- on-site or remotely
- in French or English
- individually or in groups

## Quotes available upon request

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# LINUX HARDENING




## a. Description

This intensive 1-day training teaches you how to harden a Linux server from the ground up. You'll learn to **minimize the system footprint**, **lock down user privileges**, **secure network exposure**, **and reinforce critical services**. Through hands-on exercises, you'll apply **real-world hardening** techniques used by security professionals.

## b. Educational goals

- Identify common entry points, privilege-escalation paths, and misconfigurations
- Minimize installed software and apply kernel and filesystem protections
- Configure strong authentication, secure sudo policies, and controlled user environments
- Secure network interfaces, firewall rules, and remote access pathways
- Deploy logging, auditing, and basic intrusion-detection mechanisms

## c. Prerequisites

- Being comfortable with CLI mode
- Familiarity with basic Linux administration

## d. Targeted audience

- System administrators
- Dev(Sec)Ops engineers

## e. Program

**Introduction**

- Why hardening matters?
- Understanding the attack surface
- Defense in depth
- Principle of least privilege
- Minimal install philosophy
- Hardening is role-specific
- Reference frameworks
- Baseline vs. hardening vs. compliance

**System baseline**

- Partitioning strategy
- Unnecessary packages & services
- Unused kernel modules
- Hardening the Kernel

**Account hardening**

- Mandatory Access Control (MAC)
- Account and sudo hardening
- Password and authentication policy

**Network hardening**

- Network exposure
- Firewalling
- Brute-force mitigation

**Services hardening**

- Dedicated users
- File permissions
- Service sandboxing

**Filesystem, permissions & integrity**

- SUID/SGID auditing
- Sticky bit and ACLs
- Immutable attributes
- File integrity monitoring

**Logging, auditing & monitoring**

- Kernel-level security event
- Centralized logging
- Intrusion detection concepts

**Automation & compliance validation**

- Scoring with Lynis
- Formal compliance scoring with OpenSCAP
- Hardening-as-code with Ansible

**Resources**

- Books
- Interactive platforms
- Courses & videos
- Tools